Physics of Complex Systems Homework 9

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Problem 1 (Density Matrix of a Single Qubit)**.** For a given ensemble of qubits the observables

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & 2i \\ -2i & 0 \end{pmatrix},$$

have been measured many times, giving the expectation values

$$\langle A \rangle = 2, \qquad \langle B \rangle = \frac{1}{2}, \qquad \langle C \rangle = 0.$$

(a) Determine the density matrix of the ensemble.

(b) Is the ensemble pure or mixed?

(c) If we measure σ^z, what is the probability to get $+1$?

(d) Determine the expectation values

$$\langle \sigma^x \rangle, \qquad \langle \sigma^y \rangle, \qquad \langle \sigma^z \rangle.$$

Proof. (a) Given a density matrix ρ, the expectation value of an operator $\mathbf{A}$ is given by $\langle \mathbf{A} \rangle = \mathrm{Tr}(\rho \mathbf{A})$. We implement the trace 1 and hermiticity conditions by defining our density matrix using

$$\rho = \begin{pmatrix} a & b \\ b^* & 1-a \end{pmatrix}.$$

Substituting this in, we find

$$\langle \mathbf{A} \rangle = 1 + 2a = 2$$
$$\langle \mathbf{B} \rangle = -1 + 2a + b + b^* = \frac{1}{2}$$

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$$\langle \mathbf{C} \rangle = -2ib + 2ib^* = 0$$

From the third equation, we see that $b = b^*$, i.e. b is real. From the first equation, we can see easily that $a = \frac{1}{2}$. Substituting this in to the second equation, we find that $2b = \frac{1}{2}$, or $b = \frac{1}{4}$. This leads to the density matrix

$$\rho = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} \end{pmatrix}.$$

(b) We do this using the trace criterion: We have

$$\rho^2 = \begin{pmatrix} \frac{5}{16} & \frac{1}{4} \\ \frac{1}{4} & \frac{5}{16} \end{pmatrix}.$$

Thus, the trace is

$$\mathrm{Tr}\left(\rho^2\right) = \frac{10}{16} \neq 1.$$

The ensemble is mixed.

(c) The probability is given by

$$\mathbb{P}(\sigma^z = 1) = \langle \psi | \sigma^z | \psi \rangle$$

where $|\psi\rangle$ is the vector within the 1-dimensional eigenspace $\sigma^z = 1$. In this basis, this eigenvector is given by $|\psi\rangle = (1, 0)^T$. Substituting, we find that

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2}.$$

Thus, the probability of measuring a 1 is $\frac{1}{2}$.

(d) We compute

$$\rho\sigma^z = \begin{pmatrix} \frac{1}{2} & -\frac{1}{4} \\ \frac{1}{4} & -\frac{1}{2} \end{pmatrix}.$$

Thus, we have

$$\langle \rho\sigma^z \rangle = 0.$$

This agrees with the result in (c); with a probability of $\frac{1}{2}$ of getting a 1, there would be a probability of $\frac{1}{2}$ of getting a -1. This would result in an expectation value of 0. Similarly, we have

$$\rho\sigma^x = \begin{pmatrix} \frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} \end{pmatrix}$$

with

$$\langle \rho\sigma^x \rangle = \frac{1}{2}$$

and

$$\rho\sigma^y = \begin{pmatrix} \frac{i}{4} & -\frac{i}{2} \\ \frac{i}{2} & -\frac{i}{4} \end{pmatrix}$$

with

$$\langle \rho\sigma^y \rangle = 0. \qquad \square$$

Problem 2 (Criterion for the Equivalence of Ensembles)**.** Consider a statistical ensemble $\{|\psi_i\rangle, p_i\}$ with the associated density matrix

$$\rho = \sum_{i=1}^{N} p_i |\psi_i\rangle\langle\psi_i|.$$

Here N is arbitrary (smaller or larger than the dimension of the Hilbert space) and the vectors $|\psi_i\rangle$ are pairwise different and normalized but not necessarily orthogonal. Moreover, let

$$\rho = \sum_{k=1}^{K} \lambda_k |k\rangle\langle k|$$

be the spectral decomposition of ρ with orthonormal eigenvectors $|k\rangle$ and eigenvalues λ_k.

The aim of this exercise is to work out an equivalence theorem for different ensembles with the same density matrix.

(a) Show that every vector $|\psi_i\rangle$ can be written as a linear combination of the eigenvectors $|k\rangle$, i.e.

$$|\psi_i\rangle \in \operatorname{span}\{|k\rangle\}.$$

(b) Use (a) to prove that every $|\psi_i\rangle$ can be written as

$$\sqrt{p_i}\,|\psi_i\rangle = \sum_{k=1}^{K} c_{ik}\sqrt{\lambda_k}\,|k\rangle,$$

with coefficients c_{ik} obeying

$$\sum_{i=1}^{N} c_{ik} c_{il}^* = \delta_{kl}, \qquad k, l = 1, \ldots, K.$$

(c) Because of (a) we know that $N \geq K$. For $N = K$ the result of (b) would imply that the coefficient matrix c_{ik} is unitary. Show that for $N > K$ the rectangular matrix c_{ik} can be extended to a square unitary matrix such that the relation in (b) still holds.

(d) So far we have proven that if

$$\sum_{i=1}^{N} p_i |\psi_i\rangle \langle \psi_i| = \sum_{k=1}^{K} \lambda_k |k\rangle \langle k|,$$

then there exists a unitary $N \times N$ matrix c_{ik} such that

$$\sqrt{p_i}\, |\psi_i\rangle = \sum_{k=1}^{N} c_{ik} \sqrt{\lambda_k}\, |k\rangle.$$

Prove the converse statement.

(e) Use (a)–(d) to show that

$$p_i = \sum_{k=1}^{N} T_{ik} \lambda_k,$$

(or in vector form $\vec{p} = T\vec{\lambda}$), where T is a doubly-stochastic matrix, i.e. all matrix entries are real and nonnegative and

$$\sum_{i=1}^{N} T_{ik} = \sum_{k=1}^{N} T_{ik} = 1.$$

(f) Use e.g. Mathematica to work out the explicit example

$$p_1 = p_2 = \frac{1}{2}, \qquad |\psi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad |\psi_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

with $N = K = 2$, and determine $\lambda_{1,2}$, $|\phi_{1,2}\rangle$, c_{ij}, and T_{ij}. Check the properties derived in (e).

Proof. **Brief note on interpretation:** In the sum

$$\rho = \sum_{k=1}^{K} \lambda_k\, |k\rangle\langle k|$$

the sum only contains the eigenvectors with **nonzero** eigenvalue. This is in contrast to the standard way the spectral theorem is written including the kernel. Otherwise, (a) would follow trivially from the spectral theorem, because the vectors $|k\rangle$ would span the space, and the opening statement in (c) would be false, because K would be the dimension of the Hilbert space, and it is not necessarily true that $N \geq \dim \mathcal{H}$.

(a) As from the comment, in this problem we basically seek to show that $|\psi_i\rangle$ is spanned by the eigenvectors with nonzero eigenvalue. However, we also know that the eigenvectors with nonzero eigenvalue span the kernel. Thus, we decompose

$$\mathcal{H} = \ker \rho \oplus \operatorname{im} \rho = \ker \rho \oplus (\ker \rho)^{\perp}.$$

We seek to show that $|\psi_i\rangle \in \operatorname{im} \rho$. Since this is an orthogonal decomposition, it suffices to show that $\langle \phi | \psi_i \rangle = 0$ for all $|\phi\rangle \in \ker \rho$, i.e. $|\psi_i\rangle$ is orthogonal to $\ker \rho$.

Consider $|\phi\rangle \in \ker \rho$. Then we have

$$0 = \langle \phi | \rho | \phi \rangle = \sum_{i=1}^{N} p_i \langle \phi | \psi_i \rangle \langle \psi_i | \phi \rangle = \sum_{i=1}^{N} p_i | \langle \phi | \psi_i \rangle |^2.$$

Since the terms in parentheses are all positive, it follows that $\langle \phi | \psi_i \rangle = 0$ for all $|\psi_i\rangle$. Since $|\phi\rangle$ was arbitrary, the conclusion follows.

(b) Since $|\psi_i\rangle$ is in the span of $|k\rangle$, by definition

$$|\psi_i\rangle = \sum_{k=1}^{n} a_{ik} |k\rangle .$$

By redefining the coefficients $a_{ik} = c_{ik}\sqrt{\lambda_k}/\sqrt{p_i}$, we can write

$$\sqrt{p_i}\, |\psi_i\rangle = \sum_{k=1}^{K} c_{ik} \sqrt{\lambda_k}\, |k\rangle .$$

Note that we require that $p_i \neq 0$, but this is given. Thus we need to solve the second part. First, we take the conjugate to obtain

$$\sqrt{p_i}\, \langle \psi_i| = \sum_{k=1}^{K} c_{ik}^* \sqrt{\lambda_k}\, \langle k| .$$

We substitute these in the expression from ρ to get

$$\rho = \sum_{i=1}^{N} p_i\, |\psi_i\rangle\langle\psi_i|$$

$$= \sum_{i=1}^{N} p_i \left[\frac{1}{\sqrt{p_i}} \sum_{k=1}^{K} c_{ik} \sqrt{\lambda_k} \left| k \right\rangle \right] \left[\frac{1}{\sqrt{p_i}} \sum_{l=1}^{K} c_{il}^* \sqrt{\lambda_l} \left\langle l \right| \right]$$
$$= \sum_{i=1}^{N} \sum_{k=1}^{K} \sum_{l=1}^{K} c_{ik} c_{il}^* \sqrt{\lambda_k \lambda_l} \left| k \right\rangle\!\left\langle l \right|$$

Now, since the $|k\rangle$ are orthogonal by the spectral theorem, we are allowed to compare coefficients. The off diagonal terms must vanish and we have

$$\sum_{i=1}^{N} c_{ik} c_{il}^* = \delta_{kl}$$

to obtain the density matrix.

(c) Note on the indices in c_{ik}: i goes from 1 to N and k goes from 1 to $K < N$. Thus, for each k, we have a vector in $\mathbb{C}^N$. The cs are orthonormal for different i. By the basis extension theorem, we can extend this basis to an orthonormal basis of $\mathbb{C}^N$, i.e. pick c_{ik} for $k > K$ such that c_{ik}s are orthonormal.

The basis extension theorem can be proven (easily) if we assume the axiom of choice: We order the sets by inclusion, and choose a maximal spanning set containing the basis of K vectors. Then, by the Gram-Schmidt procedure, we can orthogonalise these vectors.

Since these spaces are finite dimensional this can also be proven without the axiom of choice, but who does that anyway?

Note: In the sum

$$\sqrt{p_i} \left| \psi_i \right\rangle = \sum_{k=1}^{K} c_{ik} \sqrt{\lambda_k} \left| k \right\rangle,$$

we must extend the kets $|k\rangle$ for $k > K$ by just choosing kets within the kernel of ρ; since then $\lambda_k = 0$, this makes no difference to the sum.

(d) We simply substitute the second equation into the first

$$\sum_{i=1}^{N} p_i \left| \psi_i \right\rangle\!\left\langle \psi_i \right| = \sum_{i=1}^{N} (\sqrt{p_i} \left| \psi_i \right\rangle)(\left\langle \psi_i \right| \sqrt{p_i})$$
$$= \sum_{i=1}^{N} \left[\sum_{k=1}^{N} c_{ik} \sqrt{\lambda_k} \left| k \right\rangle \right] \left[\sum_{l=1}^{N} c_{il}^* \sqrt{\lambda_l} \left\langle l \right| \right]$$

$$= \sum_{k=1}^{N} \sum_{l=1}^{N} \underbrace{\left[\sum_{i=1}^{N} c_{ik} c_{il}^{*} \right]}_{\delta_{kl}} \sqrt{\lambda_k \lambda_l} \, |k\rangle\langle l|$$

$$= \sum_{k=1}^{N} \lambda_k \, |k\rangle\langle k|$$

(e) We take norms of the equation

$$\sqrt{p_i} \, |\psi_i\rangle = \sum_{k=1}^{n} c_{ik} \sqrt{\lambda_k} \, |k\rangle$$

to get

$$p_i = \sum_{k=1}^{N} |c_{ik}|^2 \lambda_k.$$

By inspection, we find

$$T_{ik} = |c_{ik}|^2.$$

Clearly, all matrix entries are real and nonnegative. Now, we evaluate the sums

$$\begin{aligned} \sum_{k=1}^{N} T_{ik} &= \sum_{k=1}^{N} |c_{ik}|^2 \\ &= \sum_{k=1}^{N} c_{ik} c_{ik}^{*} \\ &= \sum_{k=1}^{N} c_{ki}^{\dagger} c_{ik} \\ &= \delta_{kk} = 1 \end{aligned}$$

The other sum follows basically identically:

$$\begin{aligned} \sum_{i=1}^{N} T_{ik} &= \sum_{i=1}^{N} |c_{ik}|^2 \\ &= \sum_{i=1}^{N} c_{ik} c_{ik}^{*} \\ &= \sum_{i=1}^{N} c_{ki}^{\dagger} c_{ik} \\ &= \delta_{kk} = 1 \end{aligned}$$

(f) We compute these in Mathematica. First, we directly compute

```
In[1]:= p1 = 1/2;

In[2]:= p2 = 1/2;

In[3]:= ψ1 = {1, 1}/√2;

In[4]:= ψ2 = {0, 1};

In[5]:= ρ = p1 KroneckerProduct[ψ1, ψ1] + p2 KroneckerProduct[ψ2, ψ2]

Out[5]= {{1/4, 1/4}, {1/4, 3/4}}
```

to find

$$\begin{aligned}\rho &= p_1 \left|\psi_1\right\rangle\!\left\langle\psi_1\right| + p_2 \left|\psi_2\right\rangle\!\left\langle\psi_2\right| \\ &= \begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}\end{aligned}$$

Then, we compute the eigenvalues and eigenvectors using

```
In[6]:= {eigvals, eigvecs} = Eigensystem[ρ]

Out[6]= {{1/4 (2 + Sqrt[2]), 1/4 (2 - Sqrt[2])}, {{-1 + Sqrt[2], 1}, {-1 - Sqrt[2], 1}}}
```

leading to eigenvalues

$$\lambda_\pm = \frac{1}{4}(2 \pm \sqrt{2})$$

and associated eigenvectors

$$\left|\phi_\pm\right\rangle = \begin{pmatrix} -1 \mp \sqrt{2} \\ 1 \end{pmatrix}.$$

Although these vectors are orthogonal, they are not normalised. Thus, we divide through

```
In[7]:= eivecsnorm = #/Norm[#] & /@ eigvecs

Out[7]= {{(-1 + Sqrt[2])/Sqrt[1 + (-1 + Sqrt[2])^2], 1/Sqrt[1 + (-1 + Sqrt[2])^2]}, {(-1 - Sqrt
[2])/Sqrt[1 + (1 + Sqrt[2])^2], 1/Sqrt[1 + (1 + Sqrt[2])^2]}}
```

To compute the c_{ik}, we note that the $|k\rangle$ form an orthonormal basis by the spectral theorem. Thus, we have

$$\sqrt{p_i}\,\langle l|\psi_i\rangle = c_{il}\sqrt{\lambda_l}.$$

Thus, we compute

$$c_{ik} = \frac{\sqrt{p_i}\,\langle k|\psi_i\rangle}{\sqrt{\lambda_k}}.$$

Defining $|\phi_1\rangle = |\phi_+\rangle = |1\rangle$ and $|\phi_2\rangle = |\phi_-\rangle = |2\rangle$, we thus evaluate these using

```
In[8]:= c = FullSimplify[Table[(Sqrt[p[[i]]] Conjugate[eivecsnorm[[k]]] . ψ[[i]])/Sqrt[eigvals[[k]]],
        {i, 1, 2}, {k, 1, 2}]]

Out[8]= {{1/Sqrt[2], - 1/Sqrt[2]}, {1/Sqrt[2], 1/Sqrt[2]}}
```

The T matrix then follows easily, from $T = |c|^2$ elementwise as

$$T = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

It is obvious that T has all real and nonegative entries. It is also obvious that all rows and columns sum up to 1.

Finally, we can compute the product

$$T\vec{\lambda} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} (2+\sqrt{2})/4 \\ (2-\sqrt{2})/4 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix} = \vec{p}. \qquad \square$$